

# Grad Café Admissions Analysis • SQL Query Report

EN.605.601 Software Concepts • Module 3 Assignment • Student: Noella Formin

## Question 1: Fall 2026 Applicant Count

**Prompt:** How many entries in your database are from applicants who applied for Fall 2026?

**Result: Fall 2026 applicant count: 19,291**

**SQL Query:**  
`SELECT COUNT(*) FROM applicants WHERE term ILIKE '%Fall 2026%';`

**Explanation:** This query filters the applicants table where the term string matches 'Fall 2026' (case-insensitively via ILIKE) and computes the total count using the COUNT(\*) aggregate function.

## Question 2: International Students Percentage

**Prompt:** Among entries that provide a nationality classification, what percentage are international students? (Excluding American and Other from numerator; excluding missing values from denominator).

**Result: Percent international: 52.25%**

**SQL Query:**  
`SELECT COUNT(*) FILTER (WHERE LOWER(us_or_international) = 'international') * 100.0 / NULLIF(COUNT(*)  
FILTER (WHERE us_or_international IS NOT NULL AND TRIM(us_or_international) <> ''), 0) FROM applicants;`

**Explanation:** Calculates the ratio of applicants explicitly marked as 'International' relative to the valid denominator of all non-null, non-blank nationality entries, formatted to two decimal places.

## Question 3: Overall Score Averages (GPA, GRE Quant, GRE Verbal, GRE Analytical Writing)

**Prompt:** What are the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores of applicants who provide each metric?

**Result: Average GPA: 3.70 | Average GRE Quantitative: 162.47 | Average GRE Verbal: 159.05 | Average GRE Analytical Writing: 5.00**

**SQL Query:**  
`SELECT AVG(gpa) AS avg_gpa, AVG(gre) AS avg_gre_quant, AVG(gre_v) AS avg_gre_verbal, AVG(gre_aw) AS  
avg_gre_aw FROM applicants;`

**Explanation:** Computes arithmetic means for each academic metric independently. In PostgreSQL, standard aggregate AVG() automatically ignores NULL rows, ensuring applicants without a specific metric do not penalize that average.

#### Question 4: Average GPA of American Fall 2026 Applicants

**Prompt:** What is the average GPA of American applicants who applied for Fall 2026?

**Result: Average GPA: 3.70**

**SQL Query:**

```
SELECT AVG(gpa) FROM applicants WHERE term ILIKE '%Fall 2026%' AND LOWER(us_or_international) = 'american' AND gpa IS NOT NULL;
```

**Explanation:** Restricts rows simultaneously to Fall 2026 and American nationality classification, calculating the average GPA over all non-null entries.

#### Question 5: Fall 2025 Acceptance Percentage

**Prompt:** What percentage of Fall 2025 entries are acceptances?

**Result: Fall 2025 acceptance percentage: 46.22%**

**SQL Query:**

```
SELECT COUNT(*) FILTER (WHERE status ILIKE '%accept%') * 100.0 / NULLIF(COUNT(*), 0) FROM applicants WHERE term ILIKE '%Fall 2025%';
```

**Explanation:** Computes the acceptance rate for Fall 2025 by taking the filtered count of rows where admission status indicates acceptance divided by the total number of Fall 2025 records.

#### Question 6: Average GPA of Accepted Fall 2026 Applicants

**Prompt:** What is the average GPA of accepted applicants who applied for Fall 2026?

**Result: Average GPA: 3.70**

**SQL Query:**

```
SELECT AVG(gpa) FROM applicants WHERE term ILIKE '%Fall 2026%' AND status ILIKE '%accept%' AND gpa IS NOT NULL;
```

**Explanation:** Filters for records with Fall 2026 term and acceptance status, aggregating the GPA scores of admitted students.

#### Question 7: Johns Hopkins University Master's in Computer Science Submissions

**Prompt:** How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science (using original downloaded fields)?

**Result: Applicant count: 23**

**SQL Query:**

```
SELECT COUNT(*) FROM applicants WHERE (program ILIKE '%Johns Hopkins%' OR program ILIKE '%JHU%') AND program ILIKE '%Computer Science%' AND degree ILIKE '%master%';
```

**Explanation:** Uses pattern matching against original program and degree fields to identify JHU / Johns Hopkins University Master's in Computer Science applicants.

### Question 8: Fall 2026 Accepted PhD in CS at Elite Institutions (Original Fields)

**Prompt:** Using original downloaded fields: How many Fall 2026 entries are acceptances from applicants applying for a PhD in Computer Science at Georgetown, MIT, Stanford, or CMU?

**Result: Applicant count: 20**

**SQL Query:**  
SELECT COUNT(\*) FROM applicants WHERE term ILIKE '%Fall 2026%' AND status ILIKE '%accept%' AND degree ILIKE '%phd%' AND program ILIKE '%Computer Science%' AND (program ILIKE '%Georgetown%' OR program ILIKE '%Massachusetts Institute of Technology%' OR program ILIKE '%MIT%' OR program ILIKE '%Stanford%' OR program ILIKE '%Carnegie Mellon%');

**Explanation:** Filters simultaneously by term (Fall 2026), status (Accepted), degree (PhD), subject (Computer Science), and institution keywords within the raw program field.

### Question 9: Fall 2026 Accepted PhD in CS at Elite Institutions (LLM Canonical Fields)

**Prompt:** Repeat Question 8 using LLM-generated university and program fields instead of original fields.

**Result: Original-field count: 20 | LLM-field count: 20 | Difference: 0**

**SQL Query:**  
SELECT COUNT(\*) FROM applicants WHERE term ILIKE '%Fall 2026%' AND status ILIKE '%accept%' AND degree ILIKE '%phd%' AND llm\_generated\_program ILIKE '%Computer Science%' AND (llm\_generated\_university ILIKE '%Georgetown%' OR llm\_generated\_university ILIKE '%Massachusetts Institute of Technology%' OR llm\_generated\_university ILIKE '%MIT%' OR llm\_generated\_university ILIKE '%Stanford%' OR llm\_generated\_university ILIKE '%Carnegie Mellon%');

**Explanation:** Replaces raw string matching with standardized LLM canonical program and university columns. Because our canonical dictionary fully mapped variations of MIT, CMU, Stanford, and Georgetown, both queries matched the target cohort.

### Question 10 (Original 1): Top 5 Universities by CS Applicant Volume & Acceptance Rate

**Prompt:** What are the top 5 universities with the highest total number of Computer Science applicant submissions, and what is the acceptance rate at each?

**Result: Georgia Tech: 45 (48.89%) | Stanford: 44 (40.91%) | JHU: 41 (43.90%) | UPenn: 41 (36.59%) | NYU: 39 (48.72%)**

**SQL Query:**  
SELECT llm\_generated\_university, COUNT(\*) AS total\_applicants, COUNT(\*) FILTER (WHERE status ILIKE '%accept%') AS accepted\_count, ROUND(COUNT(\*) FILTER (WHERE status ILIKE '%accept%') \* 100.0 / COUNT(\*), 2) AS acceptance\_rate\_pct FROM applicants WHERE llm\_generated\_program ILIKE '%Computer Science%' AND llm\_generated\_university IS NOT NULL GROUP BY llm\_generated\_university ORDER BY total\_applicants DESC, acceptance\_rate\_pct DESC LIMIT 5;

**Explanation:** Aggregates applicant counts and calculates proportional acceptance percentages grouped by university for Computer Science applicants, ordering by total volume.

**Question 11 (Original 2): Fall 2026 Outcome Metrics (Accepted vs. Rejected Comparison)**

**Prompt:** For Fall 2026 applicants, how do average GPA and average GRE Quantitative scores compare between Accepted vs. Rejected applicants?

**Result: Accepted: 8,691 admits | Avg GPA = 3.70 | Avg GRE Q = 162.38 Rejected: 8,581 applicants | Avg GPA = 3.70 | Avg GRE Q = 162.49**

**SQL Query:**

```
SELECT CASE WHEN status ILIKE '%accept%' THEN 'Accepted' WHEN status ILIKE '%reject%' THEN 'Rejected' ELSE 'Other' END AS admission_outcome, COUNT(*) AS applicant_count, ROUND(AVG(gpa)::numeric, 2) AS avg_gpa, ROUND(AVG(gre)::numeric, 2) AS avg_gre_quant FROM applicants WHERE term ILIKE '%Fall 2026%' AND (status ILIKE '%accept%' OR status ILIKE '%reject%') GROUP BY admission_outcome ORDER BY admission_outcome ASC;
```

**Explanation:** Groups Fall 2026 admissions by binary decision outcome to evaluate whether self-reported scores exhibit differentiation between accepted and rejected applicants.